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Science Architecture

How the orchestrator takes an element from line list to published abundance

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FRAMEWORK // HOW THE CODEX WORKS

Science *Architecture*

How one command takes an element from a published line list to a checked, published abundance — and what each element we have run through it taught it.

The Method page explains the science: which spectra, which lines, which models, how an abundance and its error bar are measured. This page explains the machinery that does that work for every element in the same way. It is called the **orchestrator**, and the point of the project right now is for it to run an element from start to finish with as little hand-steering as possible. Each element we put through it is a test of the orchestrator as much as a measurement of the Sun.

SECTION 01

One command per element

Every element goes through the same three stages, each started by one command:

1 · PREPARE	Check everything the measurement depends on, before anything is measured. Produces one report and a READY or NOT READY answer.

2 · PUBLISH	Measure every combination of spectrum, band and model that applies, build each result's error bar, publish what passes, withdraw what it replaces, and compare the results to the literature.
3 · COVERAGE	List every combination the element should have and what happened to it, so "finished" is something the pipeline can show rather than something someone asserts.

The division of labour is deliberate. The orchestrator does the work. The person or AI agent running it starts the commands and reports what came out. Code changes only after a problem has been reported and agreed. Two decisions stay with Ryan: approving the prepare report (including any lines to drop), and merging the results into the public record.

The step-by-step procedure an operator follows is kept in the code repository as the orchestrator runbook ↗, next to the code it describes, so the two change together.

SECTION 02

Prepare: checks before measuring

Earlier elements were measured first and checked afterwards, and every late discovery cost a full rerun. Prepare moves those checks to the front. It reads files only, and it asks six questions:

CHECK	THE QUESTION
Published line set	Is the solar literature's own line list for this element registered, and does each of its lines have the oscillator strength (gf) the authors used, with their stated uncertainty?
gf sources	Is every source of atomic data classified by how its errors correlate between lines? An unclassified source would leave part of the error bar unpriced.
Lines to drop	Which lines are too strong to measure reliably *and* unused by the solar literature? These are proposals only.
NLTE coverage	Does the correction table for departures from local thermodynamic equilibrium cover every graded line?
Spectra	Does each prepared spectrum actually contain each line, at its rest wavelength, in its window?
Literature	Is there a file of published solar values to compare against, and does every reference in it resolve to a real paper?

The answer is READY or NOT READY with the blockers named. Nothing about the data is changed until the report has been read and the operator confirms it; that confirmation is the one human checkpoint before measurement. A proposed line drop comes with any literature that argues against it. Magnesium showed why: one strong line is absent from the reference line table but is discussed explicitly as a solar line in another paper, and a rule that only looked at the first source would have dropped it.

SECTION 03

The run matrix: every cell has an outcome

One element is not one measurement. The orchestrator expands it into **cells**: one cell per wavelength band × prepared spectrum × ion × line grade × model treatment. Iron in the Sun is several hundred cells.

None of those lists is typed into the code. The spectra come from the holdings registry, the band edges from one band table, each instrument's reach from the instrument catalogue, the model treatments from the model registry and the graded lines from the atomic-data table. A second hand-written copy of any of those facts is how two parts of a pipeline end up disagreeing, and that happened here more than once before the rule existed.

Every cell ends a run with exactly one outcome, and each outcome points at a different owner:

- **Done** — measured in this run; **skipped** — already current (next section).
- **Held** — a step that must come before measurement is not complete, and the reason says which.
- **Blocked** — the combination cannot be run at all, for example a model that does not exist for this element.
- **Not ready** — the data is not in a usable state for this cell.
- **Failed** — the cell ran and broke. One failure stops one cell, never the whole run.

A cell is never silently dropped. A gap that nobody wrote down and an honest "nothing here" look the same to everything downstream, and only one of them is a

finding.

SECTION 04

Work already current is not redone

A full element run takes hours, so the orchestrator records a fingerprint for every cell it builds: the cell's description, the exact code it ran, and every input file, including the spectrum itself. On the next run, a cell whose fingerprint is unchanged is skipped. A code change or a newly prepared spectrum changes the fingerprint and the cell runs again; the passing of time does not.

The same fingerprint tells the coverage report when a published value has gone **stale**: published, but built from lines, atomic data, a spectrum or code that has since changed.

SECTION 05

Measure, then build the error bar

Each runnable cell is measured the way the Method page describes, mostly by fitting a synthetic spectrum to each line. To make several hundred cells affordable, each line is synthesised once on a fine grid and the instrument's resolution is applied afterwards. Before adopting this, the cached spectrum was checked against a direct synthesis and they agreed to the fourth decimal place in abundance.

Different model treatments (one-dimensional or averaged three-dimensional atmospheres, with or without NLTE) each produce their own result. They are reported side by side and never averaged into one number.

Then comes the error bar. For each published value, every source of uncertainty is measured the same way: the measurement is repeated with *only* that one input changed — a warmer model atmosphere, a different continuum placement, the other telluric correction — and the shift is recorded. These paired reruns are most of the run time. A value whose error bar is missing any component is held rather than published with a partial error.

SECTION 06

Publish, and withdraw what is replaced

Results go into the element's published file through the same gates every product passes: a complete error bar, the right grade for its lines, and a record of where it came from. Anything that fails stays held, with the reason written down.

When a run rebuilds a cell and no longer publishes a value it published before, the old value is **withdrawn**: moved to an archive with the date and the reason, never deleted, so the record of what was once shown survives. Iron showed that the withdrawal has to be limited to the cells and model treatments the run actually rebuilt: an early run without that limit withdrew rows it had not rebuilt, and they had to be restored. The limit was added for iron, a later merge undid it in the shared code, and putting it back is open work.

SECTION 07

Check against the literature

After publishing, each result is compared with the published solar values in the element's literature file and marked as inside or outside the published range. A

result on a different scale (for example one-dimensional against a three-dimensional NLTE reference) is flagged as a scale mismatch rather than compared directly. The check produces a list of problem lines for a person to look at.

It never changes a number or a line selection. A pipeline that adjusted its results until they matched the literature would only be repeating the literature back.

SECTION 08

What “done” means

The coverage report gives every cell exactly one of four states:

- **Published** — a value with its error bar and line count is in the public record.
- **Held** — it cannot be produced, and the reason is stated.
- **Not run** — it applies and has not been measured yet: work is owed.
- **Stale** — it is published, but its inputs have changed since.

An element is done when nothing is *not run* or *stale*. Even then, a done coverage table is not taken as proof the published file is clean: the full test suite is run on the branch and on the main line, and only failures that appear on the branch alone count. Silicon is the example. Its coverage read done while five superseded products were still live, and only the full suite caught it.

SECTION 09

What the runs taught it

Each element run through the orchestrator so far has found something the orchestrator then learned to do by itself:

- **Silicon, the first real run.** The orchestrator was sending uncorrected spectra to be measured and refusing the corrected ones because of out-of-date labels. It now never uses a raw spectrum. Two parts of the code also disagreed on where the bands start and end; both now read the same band table. And five model treatments had been writing into one results file, so each now writes its own.
- **Aluminium** followed silicon through the corrected version, and the two together became the worked example the runbook tells every later element to copy.
- **Iron.** Iron's Reference Grade became the line set Asplund and collaborators used, plus every line with a laboratory-measured gf. This brought in the synthesis cache, the lesson about scoping withdrawals described above, and a rule that results from experimental models are flagged straight from the model registry instead of by hand.
- **Magnesium**, the current test. Prepare is where its problems surfaced: a line-drop rule that needed to look past one reference table, and an NLTE step that needs a different software environment from the synthesis, which the procedure now checks before starting. A review of magnesium's atomic data before release found a gf entered with the wrong sign, and its results are being rebuilt on the corrected value.

Sodium is next, and the remaining elements' reference line lists are already assembled for it.

SECTION 10

Guardrails kept from the earlier design

- **Validate, don't tune.** A model atom or correction table is never adjusted to reproduce a trusted value. The physics has to produce it; the trusted value is the test, not the target.
- **No silent fallback.** If a line has no NLTE correction, or a line's atomic data cannot be found, the result is held. It is never quietly measured another way.
- **Models are never mixed.** One-dimensional and three-dimensional, LTE and NLTE results stay separate products.
- **Spectra are prepared once.** Telluric correction and continuum normalisation are done once per spectrum and then frozen for every element.
- **Honest blanks.** A result cannot be shown as empty when a measurement exists, and a measurement cannot be shown where none exists. The pipeline refuses that contradiction rather than letting it reach the record.
- **The record is append-only.** Superseded values are archived with their reason, not erased.

SECTION 11

Where it runs

Production runs happen on one machine, Sirius, which holds the spectra and the reference software. The development Mac mirrors it. Every file location comes from one path register, so the same command finds the same data on either host and another operator can reproduce a run. The automated tests run on Sirius too. Because the main line has a few known failing tests, a change is judged by comparing its failures with main’s, not by the raw count.

DOCUMENT BOUNDARIES

This page describes how the pipeline is organised. The Method page describes the science it applies. The Glossary defines terms, and the Sources page lists every reference. Each element’s page and downloadable appendix carry its values, diagnostics and caveats.